

DeepHeal

AI-Powered Multimodal Mental Wellness Application

Professional 4-Month Development Proposal

- **Project Overview**

DeepHeal is an advanced AI-powered mental wellness mobile application designed to provide intelligent, non-clinical emotional support by analyzing user emotions through multiple input modalities including text, voice, and facial expressions. The system combines these signals to deliver accurate emotional understanding and empathetic, context-aware responses.

- **Core AI Vision**

The core concept of DeepHeal is a multimodal Artificial Intelligence system capable of understanding emotional states by simultaneously processing textual input, vocal tone, and facial expressions. These inputs are fused into a unified emotional assessment that guides the AI's response behavior.

- **Multimodal Emotion Detection**

- Text Analysis: Natural Language Processing to detect emotional tone, stress indicators, and intent.
- Voice Analysis: Acoustic feature extraction including pitch, tone, and speech patterns.
- Facial Expression Analysis: Computer vision-based detection of facial landmarks and micro-expressions.
- Emotion Fusion Engine: Combines all inputs to determine dominant emotional state.

- **Key Features**

- Secure user authentication
- Multimodal emotion input (text, voice, facial)
- AI-powered emotional support chat
- Context-aware conversation memory
- Session history and emotional trends
- Profile and privacy settings

- **Platform & Technology**

- Platforms: Android (Phase 1), iOS-ready architecture
- Framework: Flutter / React Native
- AI Models: NLP models, speech emotion recognition, facial emotion recognition
- Backend: Secure AI processing backend
- Data Security: Encrypted storage and controlled access

• **Development Timeline (4 Months)**

Month 1 – Research & System Design

Requirements finalization, AI model selection, multimodal architecture design, compliance and ethics planning.

Month 2 – UI/UX & Core App Development

UI implementation, authentication, dashboard creation, camera and microphone permissions handling.

Month 3 – AI Model Integration

Integration of text, voice, and facial emotion detection models and development of emotion fusion logic.

Month 4 – Testing, Optimization & Deployment

Performance testing, accuracy validation, security audits, final build generation, and handover.

• **Security, Privacy & Ethical Compliance**

User consent is required for microphone and camera usage. No data is used for clinical diagnosis. All emotional data is securely processed and stored in compliance with ethical AI standards.

• **Final Deliverables**

- Android application build
- Multimodal AI integration
- Complete source code
- UI/UX assets
- Technical and user documentation

• **Conclusion**

DeepHeal delivers a comprehensive AI-driven emotional understanding system using text, voice, and facial inputs, positioning it as a next-generation digital mental wellness companion.

Additional Project-Related Features which are needed to be added

1. Manual mood selection to support AI-detected emotions.
2. Simple feedback option after each therapy response.
3. Basic mood history showing recent emotional states.
4. Daily emotional check-in reminder notification.
5. Emotion intensity selection (low, medium, high).
6. Simple onboarding guide explaining app usage.